

BA 64060 Final Project

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This script:

- 1) Loads and cleans 2023 EIA-923 YTD data (Group3Data.csv)
- 2) Creates an efficiency feature (MMBtu per MWh)
- 3) Scales variables for clustering
- 4) Splits into train/test (75/25) per the assignment
- 5) Uses elbow and silhouette methods to pick k
- 6) Runs k-means with $k = 3$
- 7) Summarizes and visualizes the clusters

1. Load libraries

dplyr: data wrangling

ggplot2: plotting

cluster: silhouette calculations

factoextra: convenient cluster visualizations

```
library(dplyr) library(ggplot2) library(cluster) library(factoextra)
```

2. Load and clean the data

I'm reading in the CSV that has the last 6 YTD columns from the EIA-923 Page 1 Generation and Fuel Data sheet.

This file should be in my working directory.

```
df <- read.csv("Group3Data.csv", check.names = FALSE)
```

Rename columns to something shorter and easier to read.

```
colnames(df) <- c("total_fuel_qty", # Total Fuel Consumption Quantity (YTD) "elec_fuel_qty", # Electric Fuel Consumption Quantity (YTD) "total_fuel_mmbtu", # Total Fuel Consumption MMBtu (YTD) "elec_fuel_mmbtu", # Electric Fuel Consumption MMBtu (YTD) "net_gen_mwh", # Net Generation (MWh, YTD) "Year" # Year )
```

A lot of these values are stored as character strings with commas.

Here I strip commas and convert everything to numeric where needed.

```
df_clean <- as.data.frame( lapply(df, function(x) { if (is.character(x)) { as.numeric(gsub(",", "", x)) } else { x } }) )
```

Now I keep only:

- rows from 2023
- rows with non-missing fuel + generation
- rows where net generation is > 0 (to avoid divide-by-zero)

```
df_2023 <- df_clean %>% filter( Year == 2023, !is.na(total_fuel_mmbtu), !is.na(net_gen_mwh), net_gen_mwh > 0 )
```

Create a new engineered feature:

$$\text{efficiency_mmbtu_per_mwh} = \text{total_fuel_mmbtu} / \text{net_gen_mwh}$$

This measures how many MMBtu of fuel are used per MWh produced.

```
df_2023 <- df_2023 %>% mutate(efficiency_mmbtu_per_mwh = total_fuel_mmbtu / net_gen_mwh)
```

Quick sanity checks

```
head(df_2023) str(df_2023) summary(df_2023$efficiency_mmbtu_per_mwh)
```

3. Select variables and scale them

I'm clustering on six features:

- total_fuel_qty
- elec_fuel_qty
- total_fuel_mmbtu
- elec_fuel_mmbtu
- net_gen_mwh
- efficiency_mmbtu_per_mwh

Because these variables are on very different scales, I standardize them (mean ~ 0 , sd ~ 1). This keeps any single variable from dominating the distance calculations.

```
cluster_vars <- df_2023 %>% select( total_fuel_qty, elec_fuel_qty, total_fuel_mmbtu, elec_fuel_mmbtu,  
net_gen_mwh, efficiency_mmbtu_per_mwh )  
cluster_scaled <- scale(cluster_vars)
```

Confirm scaling: means should be ~ 0 and sds should be ~ 1

```
colMeans(cluster_scaled) apply(cluster_scaled, 2, sd)
```

4. Train/Test split (75% / 25%)

Even though this is unsupervised, the assignment requires a split, so I randomly assign 75% of rows to a “train” set and 25% to “test”.

```
set.seed(123) # for reproducibility
n <- nrow(cluster_scaled) train_idx <- sample(1:n, size = 0.75 * n)
train <- cluster_scaled[train_idx, ] test <- cluster_scaled[-train_idx, ]
dim(train) # training dimensions dim(test) # test dimensions
```

5. Elbow method to choose k

I compute total within-cluster sum of squares (WSS) for $k = 1$ to 10.

The “elbow” is the point where adding more clusters stops giving big decreases in WSS.

```
wss <- sapply(1:10, function(k) { kmeans(train, centers = k, nstart = 20)$tot.withinss })
```

Plot WSS vs k so I can visually inspect the elbow.

```
plot( 1:10, wss, type = “b”, xlab = “Number of Clusters (k)”, ylab = “Total Within-Cluster Sum of Squares (WSS)”, main = “Elbow Method for Choosing k” )
```

6. Silhouette method to choose k

I also compute the average silhouette width for $k = 2$ to 10.

Higher silhouette values indicate better cluster separation and cohesion.

```
avg_sil <- sapply(2:10, function(k) { km <- kmeans(train, centers = k, nstart = 20) ss <- silhouette(km$cluster, dist(train)) mean(ss[, 3]) })
```

Plot average silhouette width by k.

```
plot( 2:10, avg_sil, type = "b", xlab = "Number of Clusters (k)", ylab = "Average Silhouette Width", main = "Silhouette Method for Choosing k" )
```

Inspect the silhouette values to see which k performs best.

```
avg_sil
```

Based on the elbow curve and silhouette pattern, I choose:

```
k_opt <- 3
```

7. Final k-means model with k = 3

I run k-means on the training data using k = 3 clusters and multiple random starts (nstart = 25) for stability.

```
set.seed(123) km_final <- kmeans(train, centers = k_opt, nstart = 25)
```

Cluster sizes show how many units fall into each group.

```
km_final$size
```

Cluster centers summarize the average standardized profile for each cluster.

```
km_final$centers
```

8. Attach clusters back to original data

I take the original (unscaled) df_2023 rows corresponding to the train indices and add the cluster assignments. This lets me interpret

the clusters in real-world units (MMBtu, MWh, etc.).

```
train_original <- df_2023[train_idx, ] train_original$cluster <- km_final$cluster
```

```
head(train_original)
```

9. Summarize each cluster

Here I create a summary table by cluster that includes:

- Number of units
- Average total fuel MMBtu
- Average net generation (MWh)
- Average efficiency (MMBtu per MWh)

This is what I would talk through in my presentation as

“what each cluster actually represents.”

```
cluster_summary <- train_original %>% group_by(cluster) %>% summarise( n_units = n(),  
  avg_total_fuel_mmbtu = mean(total_fuel_mmbtu, na.rm = TRUE), avg_net_gen_mwh = mean(net_gen_mwh,  
  na.rm = TRUE), avg_efficiency_mmbtu_mwh = mean(efficiency_mmbtu_per_mwh, na.rm = TRUE),  
  .groups = "drop" )  
cluster_summary
```

10. Visualize clusters in reduced space

This plot uses factoextra to show the clusters in a reduced

2D space. Each point is a plant/unit and colored by its

assigned cluster. The convex hulls make the grouping clearer.

```
fviz_cluster( list(data = train, cluster = km_final$cluster), geom = "point", ellipse.type = "convex", main  
= "K-Means Clustering of Power Units (k = 3)" )
```

11. Visualize efficiency vs. generation by cluster

This scatter plot compares net generation (MWh) versus efficiency (MMBtu per MWh). Coloring by cluster lets me see which groups are:

- high output and high intensity,
- lower output and less efficient,
- or more balanced in terms of fuel use per MWh.

```
ggplot( train_original, aes( x = net_gen_mwh, y = efficiency_mmbtu_per_mwh, color = factor(cluster)
) ) + geom_point(alpha = 0.5) + labs( title = "Clusters by Net Generation and Fuel Efficiency", x = "Net
Generation (MWh)", y = "Efficiency (MMBtu per MWh)", color = "Cluster" )
```